



平面向量专项练习题

题型 1: 平行与垂直(Parallel and Perpendicular)

题目 1-1(Question 1-1)

已知向量 $\vec{a} = (-4, 3)$, $\vec{b} = (6, m)$, 且 $\vec{a} \perp \vec{b}$, 则 $m =$ ____.

Given vectors $\vec{a} = (-4, 3)$ and $\vec{b} = (6, m)$, and $\vec{a} \perp \vec{b}$, then $m =$ ____.

题目 1-2(Question 1-2)

已知向量 $\vec{a} = (2, 6)$, $\vec{b} = (-1, \lambda)$, 若 $\vec{a} \parallel \vec{b}$, 则 $\lambda =$ ____.

Given vectors $\vec{a} = (2, 6)$ and $\vec{b} = (-1, \lambda)$. If $\vec{a} \parallel \vec{b}$, then $\lambda =$ ____.

题目 1-3(Question 1-3)

设向量 $\vec{a} = (1, 0)$, $\vec{b} = (-1, m)$. 若 $\vec{a} \perp (m\vec{a} - \vec{b})$, 则 $m =$ ____.

Let vectors $\vec{a} = (1, 0)$ and $\vec{b} = (-1, m)$. If $\vec{a} \perp (m\vec{a} - \vec{b})$, then $m =$ ____.

题目 1-4(Question 1-4)

已知向量 $\vec{a} = (1, 2)$, $\vec{b} = (2, -2)$, $\vec{c} = (1, \lambda)$. 若 $\vec{c} \parallel (2\vec{a} + \vec{b})$, 则 $\lambda =$ ____.

Given vectors $\vec{a} = (1, 2)$, $\vec{b} = (2, -2)$, and $\vec{c} = (1, \lambda)$. If $\vec{c} \parallel (2\vec{a} + \vec{b})$, then $\lambda =$ ____.

题目 1-5(2024·新高考 1 卷)(Question 1-5 (2024·New College Entrance Examination Volume 1))

已知向量 $\vec{a} = (0, 1)$, $\vec{b} = (2, x)$, 若 $\vec{b} \perp (\vec{b} - 4\vec{a})$, 则 $x =$ ()

Given vectors $\vec{a} = (0, 1)$ and $\vec{b} = (2, x)$. If $\vec{b} \perp (\vec{b} - 4\vec{a})$, then $x =$ ()

A. -2 B. -1 C. 1 D. 2

题目 1-6(2024·甲卷)(Question 1-6 (2024·National Examination Volume A))

已知向量 $\vec{a} = (x + 1, x)$, $\vec{b} = (x, 2)$, 则()

Given vectors $\vec{a} = (x + 1, x)$ and $\vec{b} = (x, 2)$, then ()

- A. “ $\vec{a} \perp \vec{b}$ ”的必要条件是“ $x = -3$ ”(The necessary condition for “ $\vec{a} \perp \vec{b}$ ” is “ $x = -3$ ”)
- B. “ $\vec{a} \parallel \vec{b}$ ”的必要条件是“ $x = -3$ ”(The necessary condition for “ $\vec{a} \parallel \vec{b}$ ” is “ $x = -3$ ”)
- C. “ $\vec{a} \perp \vec{b}$ ”的充分条件是“ $x = 0$ ”(The sufficient condition for “ $\vec{a} \perp \vec{b}$ ” is “ $x = 0$ ”)
- D. “ $\vec{a} \parallel \vec{b}$ ”的充分条件是“ $x = -1 + \sqrt{3}$ ”(The sufficient condition for “ $\vec{a} \parallel \vec{b}$ ” is “ $x = -1 + \sqrt{3}$ ”)

题型 2: 线性表示(Linear Representation)

题目 2-1(2022·新高考)(Question 2-1 (2022·New College Entrance Examination))

在 $\triangle ABC$ 中,点 D 在边 AB 上, $BD = 2DA$ 。记 $\overrightarrow{CA} = \vec{m}$, $\overrightarrow{CD} = \vec{n}$, 则 $\overrightarrow{CB} =$ ()

In $\triangle ABC$, point D is on side AB with $BD = 2DA$. Let $\overrightarrow{CA} = \vec{m}$ and $\overrightarrow{CD} = \vec{n}$, then $\overrightarrow{CB} =$ ()

- A. $3\vec{m} - 2\vec{n}$ B. $-2\vec{m} + 3\vec{n}$ C. $3\vec{m} + 2\vec{n}$ D. $2\vec{m} + 3\vec{n}$

题目 2-2(Question 2-2)

在 $\triangle ABC$ 中, AD 为 BC 边上的中线, E 为 AD 的中点, 则 $\overrightarrow{EB} =$ ()

In $\triangle ABC$, AD is the median on side BC , and E is the midpoint of AD , then $\overrightarrow{EB} =$ ()

- A. $\frac{3}{4}\overrightarrow{AB} - \frac{1}{4}\overrightarrow{AC}$ B. $\frac{1}{4}\overrightarrow{AB} - \frac{3}{4}\overrightarrow{AC}$ C. $\frac{3}{4}\overrightarrow{AB} + \frac{1}{4}\overrightarrow{AC}$ D. $\frac{1}{4}\overrightarrow{AB} + \frac{3}{4}\overrightarrow{AC}$

题目 2-3(Question 2-3)

在 $\triangle ABC$ 中, D 是 AB 边上的中点, 则 $\overrightarrow{CB} =$ ()

In $\triangle ABC$, D is the midpoint of side AB , then $\overrightarrow{CB} =$ ()

- A. $2\overrightarrow{CD} + \overrightarrow{CA}$ B. $\overrightarrow{CD} - 2\overrightarrow{CA}$ C. $2\overrightarrow{CD} - \overrightarrow{CA}$ D. $\overrightarrow{CD} + 2\overrightarrow{CA}$

题目 2-4(Question 2-4)

已知点 D 在 $\triangle ABC$ 的边 AC 上, $CD = 2DA$, 点 E 是 BD 中点, 则 $\overrightarrow{EC} =$ ()

Given that point D is on side AC of $\triangle ABC$ with $CD = 2DA$, and point E is the midpoint of



BD , then $\overrightarrow{EC} = ()$

- A. $\frac{1}{3}\overrightarrow{AB} + \frac{2}{3}\overrightarrow{AC}$ B. $\frac{1}{2}\overrightarrow{AB} + \frac{2}{3}\overrightarrow{AC}$ C. $-\frac{1}{3}\overrightarrow{AB} + \frac{5}{6}\overrightarrow{AC}$ D. $-\frac{1}{2}\overrightarrow{AB} + \frac{5}{6}\overrightarrow{AC}$

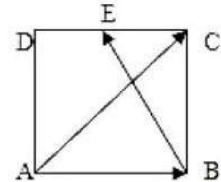
题目 2-5(Question 2-5)

如图, $ABCD$ 为正方形, E 是 CD 的中点, 且 $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AC} = \vec{b}$, 则 $\overrightarrow{BE}$ 等于()

As shown in the figure, $ABCD$ is a square, E is the midpoint of CD ,

and $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AC} = \vec{b}$, then $\overrightarrow{BE}$ is equal to ()

- A. $\vec{b} + \frac{1}{2}\vec{a}$ B. $\vec{b} - \frac{3}{2}\vec{a}$ C. $\vec{a} + \frac{1}{2}\vec{b}$ D. $\vec{a} - \frac{3}{2}\vec{b}$



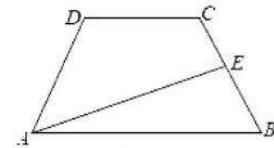
题目 2-6(Question 2-6)

如图, 梯形 $ABCD$ 中, $AB \parallel CD$, $AB = 2CD$, E 为 BC 中点, 则 $\overrightarrow{AE} = ()$

As shown in the figure, in trapezoid $ABCD$, $AB \parallel CD$ and $AB = 2CD$, E is the midpoint of

BC , then $\overrightarrow{AE} = ()$

- A. $\frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AD}$ B. $\frac{3}{4}\overrightarrow{AB} + \overrightarrow{AD}$ C. $\frac{3}{4}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AD}$ D. $\frac{3}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AD}$



题目 2-7(Question 2-7)

在 $\triangle ABC$ 中, D 是 BC 边的中点, $\overrightarrow{AE} = 2\overrightarrow{EB}$, CE 与 AD 交于点 F , 则 $\overrightarrow{AF} = ()$

In $\triangle ABC$, D is the midpoint of side BC , $\overrightarrow{AE} = 2\overrightarrow{EB}$, and CE intersects AD at point F ,

then $\overrightarrow{AF} = ()$

- A. $\frac{2}{3}\overrightarrow{AB} + \frac{2}{3}\overrightarrow{AC}$ B. $\frac{1}{3}\overrightarrow{AB} + \frac{3}{4}\overrightarrow{AC}$ C. $\frac{2}{5}\overrightarrow{AB} + \frac{2}{5}\overrightarrow{AC}$ D. $\frac{2}{9}\overrightarrow{AB} + \frac{2}{9}\overrightarrow{AC}$

题目 2-8(Question 2-8)

在平行四边形 $ABCD$ 中, E , F 分别为 CD , BC 的中点, 则 $\overrightarrow{AE} = ()$

In parallelogram $ABCD$, E and F are the midpoints of CD and BC respectively, then

$\overrightarrow{AE} = ()$

- A. $\frac{3}{4}\overrightarrow{AD} + \frac{1}{2}\overrightarrow{AF}$ B. $\frac{1}{2}\overrightarrow{AD} + \frac{1}{2}\overrightarrow{AF}$ C. $\frac{1}{2}\overrightarrow{AD} + \frac{3}{4}\overrightarrow{AF}$ D. $\overrightarrow{AD} + \frac{1}{2}\overrightarrow{AF}$

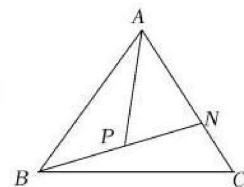
题目 2-9(Question 2-9)

如图,在 $\triangle ABC$ 中, $\overrightarrow{AN} = 2\overrightarrow{NC}$, P 是 BN 上一点,若 $\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$, 则实数 t 的值为 ()

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AN} = 2\overrightarrow{NC}$, P is a point on BN . If

$\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$, then the value of the real number t is ()

- A. $\frac{1}{6}$ B. $\frac{2}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$



题目 2-10(Question 2-10)

在 $\triangle ABC$ 中,点 M 、 N 分别在边 BC 、 CA 上,若 $\overrightarrow{BC} = 2\overrightarrow{BM}$, $\overrightarrow{CA} = 3\overrightarrow{CN}$, 则 $\overrightarrow{MN} =$ ()

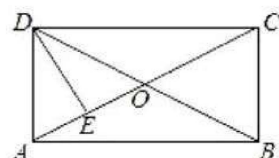
In $\triangle ABC$, points M and N are on sides BC and CA respectively. If $\overrightarrow{BC} = 2\overrightarrow{BM}$ and $\overrightarrow{CA} = 3\overrightarrow{CN}$, then $\overrightarrow{MN} =$ ()

- A. $-\frac{1}{2}\overrightarrow{AB} + \frac{1}{6}\overrightarrow{AC}$ B. $\frac{1}{2}\overrightarrow{AB} - \frac{1}{6}\overrightarrow{AC}$ C. $\frac{1}{6}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{AC}$ D. $\frac{1}{6}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AC}$

题目 2-11(Question 2-1)

如图所示,矩形 $ABCD$ 的对角线相交于点 O , E 为 AO 的中点,若

$\overrightarrow{DE} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$), 则 $\lambda \cdot \mu$ 等于 ()



As shown in the figure, the diagonals of rectangle $ABCD$ intersect at

point O , E is the midpoint of AO . If $\overrightarrow{DE} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$), then $\lambda \cdot \mu$ is equal to ()

- A. $-\frac{3}{16}$ B. $\frac{3}{16}$ C. $\frac{1}{2}$ D. $-\frac{1}{2}$

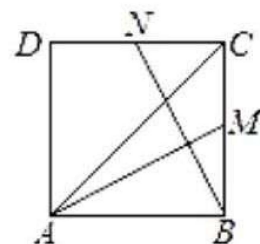
题目 2-12(Question 2-12)

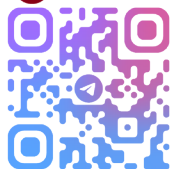
如图,正方形 $ABCD$ 中, M 、 N 分别是 BC 、 CD 的中点,若 $\overrightarrow{AC} = \lambda\overrightarrow{AM} + \mu\overrightarrow{BN}$, 则 $\lambda + \mu =$

()

As shown in the figure, in square $ABCD$, M and N are the midpoints

of BC and CD respectively. If $\overrightarrow{AC} = \lambda\overrightarrow{AM} + \mu\overrightarrow{BN}$, then $\lambda + \mu =$ ()





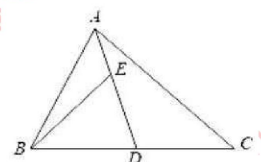
A. 2 B. $\frac{8}{3}$ C. $\frac{6}{5}$ D. $\frac{8}{5}$

题目 2-13(Question 2-13)

如图所示, $\triangle ABC$ 中, 点 D 是线段 BC 的中点, E 是线段 AD 的靠近 A 的三等分点, 则

$\overrightarrow{AC} = ()$

As shown in the figure, in $\triangle ABC$, point D is the midpoint of segment BC , and E is the trisection point of segment AD close to A , then



$\overrightarrow{AC} = ()$

A. $\frac{4}{3}\overrightarrow{AD} + \overrightarrow{BE}$ B. $\frac{5}{3}\overrightarrow{AD} + \overrightarrow{BE}$ C. $\frac{4}{3}\overrightarrow{AD} + \frac{1}{2}\overrightarrow{BE}$ D. $\frac{5}{3}\overrightarrow{AD} + \frac{1}{2}\overrightarrow{BE}$

题型 3: 数量积(Dot Product)

题目 3-1(Question 3-1)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 1$, $|\vec{b}| = \sqrt{3}$, $|\vec{a} - 2\vec{b}| = 3$, 则 $\vec{a} \cdot \vec{b} = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 1$, $|\vec{b}| = \sqrt{3}$, and $|\vec{a} - 2\vec{b}| = 3$, then $\vec{a} \cdot \vec{b} = ()$

A. -2 B. -1 C. 1 D. 2

题目 3-2(Question 3-2)

已知 $\overrightarrow{AB} = (2, 3)$, $\overrightarrow{AC} = (3, t)$, $|\overrightarrow{BC}| = 1$, 则 $\overrightarrow{AB} \cdot \overrightarrow{BC} = ()$

Given $\overrightarrow{AB} = (2, 3)$, $\overrightarrow{AC} = (3, t)$, and $|\overrightarrow{BC}| = 1$, then $\overrightarrow{AB} \cdot \overrightarrow{BC} = ()$

A. -3 B. -2 C. 2 D. 3

题目 3-3(Question 3-3)

正方形 $ABCD$ 的边长是 2, E 是 AB 的中点, 则 $\overrightarrow{EC} \cdot \overrightarrow{ED} = ()$

The side length of square $ABCD$ is 2, and E is the midpoint of AB , then $\overrightarrow{EC} \cdot \overrightarrow{ED} = ()$

A. $\sqrt{5}$ B. 3 C. $2\sqrt{5}$ D. 5

题目 3-4(Question 3-4)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 1$, $\vec{a} \cdot \vec{b} = -1$, 则 $\vec{a} \cdot (2\vec{a} - \vec{b}) = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 1$ and $\vec{a} \cdot \vec{b} = -1$, then $\vec{a} \cdot (2\vec{a} - \vec{b}) = ()$

A. 4 B. 3 C. 2 D. 0

题目 3-5(Question 3-5)

设向量 $\vec{a}$, $\vec{b}$ 的夹角的余弦值为 $\frac{1}{3}$, 且 $|\vec{a}| = 1$, $|\vec{b}| = 3$, 则 $(2\vec{a} + \vec{b}) \cdot \vec{b} = \underline{\hspace{2cm}}$.

Let the cosine value of the angle between vectors $\vec{a}$ and $\vec{b}$ be $\frac{1}{3}$, and $|\vec{a}| = 1$, $|\vec{b}| = 3$, then $(2\vec{a} + \vec{b}) \cdot \vec{b} = \underline{\hspace{2cm}}$.

题目 3-6(Question 3-6)

三角形 ABC 中, D 是 BC 中点, $AB = 2$, $BC = 3$, $AC = 4$, 则 $\overrightarrow{AD} \cdot \overrightarrow{AB} = \underline{\hspace{2cm}}$.

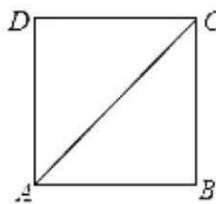
In triangle ABC , D is the midpoint of BC , $AB = 2$, $BC = 3$, $AC = 4$, then $\overrightarrow{AD} \cdot \overrightarrow{AB} = \underline{\hspace{2cm}}$.

题目 3-7(Question 3-7)

如图,正方形 $ABCD$ 的边长为 3, 求 $\overrightarrow{AB} \cdot \overrightarrow{AC} = \underline{\hspace{2cm}}$.

As shown in the figure, the side length of square $ABCD$ is 3, find $\overrightarrow{AB} \cdot \overrightarrow{AC}$.

$\overrightarrow{AC} = \underline{\hspace{2cm}}$.



题型 4: 模长(Magnitude)

题目 4-1(2024·新高考 II)(Question 4-1 (2024·New College Entrance Examination Volume II))

已知向量 $\vec{a}$, $\vec{b}$ 满足: $|\vec{a}| = 1$, $|\vec{a} + 2\vec{b}| = 2$, 且 $(\vec{b} - 2\vec{a}) \perp \vec{b}$, 则 $|\vec{b}| = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying: $|\vec{a}| = 1$, $|\vec{a} + 2\vec{b}| = 2$, and $(\vec{b} - 2\vec{a}) \perp \vec{b}$, then $|\vec{b}| = ()$

A. $\frac{1}{2}$ B. $\frac{\sqrt{2}}{2}$ C. $\frac{\sqrt{3}}{2}$ D. 1



题目 4-2(Question 4-2)

已知向量 $\vec{a} = (2,1)$, $\vec{b} = (-2,4)$,则 $|\vec{a} - \vec{b}| = ()$

Given vectors $\vec{a} = (2,1)$ and $\vec{b} = (-2,4)$, then $|\vec{a} - \vec{b}| = ()$

A. 2 B. 3 C. 4 D. 5

题目 4-3(Question 4-3)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $\vec{a} + \vec{b} = (2,3)$, $\vec{a} - \vec{b} = (-2,1)$,则 $|\vec{a}|^2 - |\vec{b}|^2 = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $\vec{a} + \vec{b} = (2,3)$ and $\vec{a} - \vec{b} = (-2,1)$, then $|\vec{a}|^2 - |\vec{b}|^2 = ()$

A. -2 B. -1 C. 0 D. 1

题目 4-4(Question 4-4)

已知向量 $\vec{a} = (2,3)$, $\vec{b} = (3,2)$,则 $|\vec{a} - \vec{b}| = ()$

Given vectors $\vec{a} = (2,3)$ and $\vec{b} = (3,2)$, then $|\vec{a} - \vec{b}| = ()$

A. $\sqrt{2}$ B. 2 C. $5\sqrt{2}$ D. 50

题目 4-5(Question 4-5)

设 $\vec{a}$, $\vec{b}$ 为单位向量,且 $|\vec{a} + \vec{b}| = 1$,则 $|\vec{a} - \vec{b}| = \underline{\hspace{2cm}}$.

Let $\vec{a}$ and $\vec{b}$ be unit vectors, and $|\vec{a} + \vec{b}| = 1$, then $|\vec{a} - \vec{b}| = \underline{\hspace{2cm}}$.

题目 4-6(Question 4-6)

设非零向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$,则()

Let non-zero vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then ()

A. $\vec{a} \perp \vec{b}$ B. $|\vec{a}| = |\vec{b}|$ C. $\vec{a} \parallel \vec{b}$ D. $|\vec{a}| > |\vec{b}|$

题目 4-7(Question 4-7)

若向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 3$, $|\vec{a} - \vec{b}| = 5$, $\vec{a} \cdot \vec{b} = 1$,则 $|\vec{b}| = \underline{\hspace{2cm}}$.

If vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 3$, $|\vec{a} - \vec{b}| = 5$,and $\vec{a} \cdot \vec{b} = 1$,then $|\vec{b}| = \underline{\hspace{2cm}}$.

题目 4-8(Question 4-8)

已知向量 $\vec{a}$, $\vec{b}$ 的夹角为 30° , $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 3$,则 $|2\vec{a} + \vec{b}| = ()$

Given the angle between vectors $\vec{a}$ and $\vec{b}$ is 30° , $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 3$,then $|2\vec{a} + \vec{b}| = ()$

A. $\sqrt{5}$ B. $\sqrt{6}$ C. $\sqrt{39}$ D. 7

题目 4-9(Question 4-9)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 1$, $|\vec{b}| = 2$,且 $\vec{a} + \vec{b} = (2\sqrt{2}, 1)$,则 $|3\vec{a} + \vec{b}| = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 1$, $|\vec{b}| = 2$,and $\vec{a} + \vec{b} = (2\sqrt{2}, 1)$,then $|3\vec{a} + \vec{b}| = ()$

A. $2\sqrt{2}$ B. $\sqrt{15}$ C. $3\sqrt{2}$ D. 5

题型 5: 夹角(Angle)

题目 5-1(Question 5-1)

设向量 $\vec{a} = (\sqrt{3}, 1)$, $\vec{b} = (-\sqrt{3}, 1)$,则 $\vec{a}$ 和 $\vec{b}$ 的夹角为()

Let vectors $\vec{a} = (\sqrt{3}, 1)$ and $\vec{b} = (-\sqrt{3}, 1)$, then the angle between $\vec{a}$ and $\vec{b}$ is ()

A. 30° B. 60° C. 120° D. 150°

题目 5-2(Question 5-2)

已知向量 $\vec{a} = (2, 2)$, $\vec{b} = (-8, 6)$,则 $\cos \langle \vec{a}, \vec{b} \rangle = \underline{\hspace{2cm}}$.

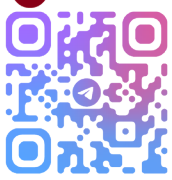
Given vectors $\vec{a} = (2, 2)$ and $\vec{b} = (-8, 6)$, then $\cos \langle \vec{a}, \vec{b} \rangle = \underline{\hspace{2cm}}$.

题目 5-3(Question 5-3)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 5$, $|\vec{b}| = 6$, $\vec{a} \cdot \vec{b} = -6$,则 $\cos \langle \vec{a}, \vec{a} + \vec{b} \rangle = ()$

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 5$, $|\vec{b}| = 6$,and $\vec{a} \cdot \vec{b} = -6$,then $\cos \langle \vec{a}, \vec{a} + \vec{b} \rangle = ()$

A. $-\frac{31}{35}$ B. $-\frac{19}{35}$ C. $\frac{17}{35}$ D. $\frac{19}{35}$



题目 5-4(Question 5-4)

已知向量 $\vec{a} = (3, 4)$, $\vec{b} = (1, 0)$, $\vec{c} = \vec{a} + t\vec{b}$, 若 $\langle \vec{a}, \vec{c} \rangle = \langle \vec{b}, \vec{c} \rangle$, 则 $t = ()$

Given vectors $\vec{a} = (3, 4)$, $\vec{b} = (1, 0)$, $\vec{c} = \vec{a} + t\vec{b}$, if $\langle \vec{a}, \vec{c} \rangle = \langle \vec{b}, \vec{c} \rangle$, then $t = ()$

A. -6 B. -5 C. 5 D. 6

题目 5-5(Question 5-5)

已知非零向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 2|\vec{b}|$, 且 $(\vec{a} - \vec{b}) \perp \vec{b}$, 则 $\vec{a}$ 与 $\vec{b}$ 的夹角为()

Given non-zero vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 2|\vec{b}|$ and $(\vec{a} - \vec{b}) \perp \vec{b}$, then the angle between $\vec{a}$ and $\vec{b}$ is ()

A. $\frac{\pi}{6}$ B. $\frac{\pi}{3}$ C. $\frac{2\pi}{3}$ D. $\frac{5\pi}{6}$

题目 5-6(Question 5-6)

已知 $\vec{a}$, $\vec{b}$ 为单位向量, 且 $\vec{a} \cdot \vec{b} = 0$, 若 $\vec{c} = 2\vec{a} - \sqrt{5}\vec{b}$, 则 $\cos \langle \vec{a}, \vec{c} \rangle = \underline{\hspace{2cm}}$.

Given $\vec{a}$ and $\vec{b}$ are unit vectors, and $\vec{a} \cdot \vec{b} = 0$, if $\vec{c} = 2\vec{a} - \sqrt{5}\vec{b}$, then $\cos \langle \vec{a}, \vec{c} \rangle = \underline{\hspace{2cm}}$.

题目 5-7(Question 5-7)

已知 $\vec{a}$ 和 $\vec{b}$ 为非零向量, 且 $|2\vec{a} + 3\vec{b}| = |2\vec{a} - 3\vec{b}|$, 则 $\vec{a}$ 与 $\vec{b}$ 的夹角为()

Given $\vec{a}$ and $\vec{b}$ are non-zero vectors, and $|2\vec{a} + 3\vec{b}| = |2\vec{a} - 3\vec{b}|$, then the angle between $\vec{a}$ and $\vec{b}$ is ()

A. $\frac{\pi}{6}$ B. $\frac{\pi}{4}$ C. $\frac{\pi}{2}$ D. $\frac{2\pi}{3}$

题目 5-8(Question 5-8)

已知 $\vec{a}$ 和 $\vec{b}$ 为非零向量, 且 $|2\vec{a} + \vec{b}| = |2\vec{a} - \vec{b}|$, 则 $\vec{a}$ 与 $\vec{b}$ 的夹角为()

Given $\vec{a}$ and $\vec{b}$ are non-zero vectors, and $|2\vec{a} + \vec{b}| = |2\vec{a} - \vec{b}|$, then the angle between $\vec{a}$ and $\vec{b}$ is ()

A. $\frac{\pi}{6}$ B. $\frac{\pi}{4}$ C. $\frac{\pi}{2}$ D. $\frac{2\pi}{3}$

题目 5-9(Question 5-9)

已知 $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| = \frac{2\sqrt{3}}{3}|\vec{a}|$, 则向量 $\vec{a} + \vec{b}$ 与 $\vec{a}$ 的夹角为()

Given $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| = \frac{2\sqrt{3}}{3}|\vec{a}|$, then the angle between vector $\vec{a} + \vec{b}$ and $\vec{a}$ is ()

- A. $\frac{5\pi}{6}$ B. $\frac{2\pi}{3}$ C. $\frac{\pi}{3}$ D. $\frac{\pi}{6}$

题目 5-10(Question 5-10)

非零向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| = 2|\vec{a}|$, 则 $\vec{a} - \vec{b}$ 与 $\vec{a}$ 的夹角为()

Non-zero vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| = 2|\vec{a}|$, then the angle between $\vec{a} - \vec{b}$ and $\vec{a}$ is ()

- A. $\frac{\pi}{6}$ B. $\frac{\pi}{3}$ C. $\frac{2\pi}{3}$ D. $\frac{5\pi}{6}$

题目 5-11(Question 5-11)

若向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 1$, $|\vec{b}| = 2$, $\vec{a} \cdot (2\vec{a} + 3\vec{b}) = 5$, 则 $\vec{a}$ 与 $\vec{b}$ 的夹角为()

If vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 1$, $|\vec{b}| = 2$, and $\vec{a} \cdot (2\vec{a} + 3\vec{b}) = 5$, then the angle between $\vec{a}$ and $\vec{b}$ is ()

- A. $\frac{\pi}{6}$ B. $\frac{\pi}{3}$ C. $\frac{2\pi}{3}$ D. $\frac{5\pi}{6}$

题目 5-12(Question 5-12)

已知平面向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{a}| = 4$, $|\vec{b}| = 1$, $(\vec{a} - \vec{b}) \perp \vec{b}$, 则 $\cos \langle \vec{a}, \vec{b} \rangle =$ ()

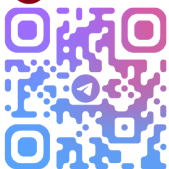
Given plane vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{a}| = 4$, $|\vec{b}| = 1$, and $(\vec{a} - \vec{b}) \perp \vec{b}$, then $\cos \langle \vec{a}, \vec{b} \rangle =$ ()

- A. $\frac{1}{4}$ B. $\frac{\sqrt{3}}{4}$ C. $\frac{\sqrt{7}}{4}$ D. $\frac{\sqrt{15}}{4}$

题目 5-13(Question 5-13)

已知非零向量 $\vec{a}$, $\vec{b}$ 满足 $(\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$, $|\vec{a} + \vec{b}| = \sqrt{3}$, 则向量 $\vec{a}$, $\vec{b}$ 的夹角是()

Given non-zero vectors $\vec{a}$ and $\vec{b}$ satisfying $(\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$ and $|\vec{a} + \vec{b}| = \sqrt{3}$, then the



angle between vectors $\vec{a}$ and $\vec{b}$ is ()

- A. $\frac{\pi}{6}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{2}$ D. $\frac{2\pi}{3}$

题目 5-14(Question 5-14)

已知 $|\vec{a}| = 1$, $|\vec{b}| = 2$, $(\vec{a} + \vec{b}) \perp \vec{a}$, 则 $\vec{a}$ 与 $\vec{b}$ 的夹角余弦值大小为()

Given $|\vec{a}| = 1$, $|\vec{b}| = 2$, and $(\vec{a} + \vec{b}) \perp \vec{a}$, then the cosine value of the angle between $\vec{a}$ and $\vec{b}$ is ()

- A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. $-\frac{\sqrt{3}}{2}$ D. $\frac{\sqrt{3}}{2}$

题型 6: 投影(Projection)

题目 6-1(Question 6-1)

已知 $|\vec{a}| = 2$, $\vec{b}$ 在 $\vec{a}$ 上的投影为 1, 则 $\vec{a} + \vec{b}$ 在 $\vec{a}$ 上的投影为()

Given $|\vec{a}| = 2$ and the projection of $\vec{b}$ on $\vec{a}$ is 1, then the projection of $\vec{a} + \vec{b}$ on $\vec{a}$ is ()

- A. -1 B. 2 C. 3 D. $\sqrt{2}$

题目 6-2(Question 6-2)

已知向量 $\vec{a}$, $\vec{b}$ 满足 $|\vec{b}| = 1$, $\vec{a} \perp \vec{b}$, 则 $\vec{a} - 2\vec{b}$ 在 $\vec{b}$ 方向上的投影向量为()

Given vectors $\vec{a}$ and $\vec{b}$ satisfying $|\vec{b}| = 1$ and $\vec{a} \perp \vec{b}$, then the projection vector of $\vec{a} - 2\vec{b}$ in the direction of $\vec{b}$ is ()

- A. 2 B. $2\vec{a}$ C. $-2\vec{b}$ D. -2

题目 6-3(Question 6-3)

已知 $\vec{a} = (2, 1)$, $\vec{b} = (-3, 4)$, 则向量 $\vec{a}$ 在向量 $\vec{b}$ 上的投影向量为()

Given $\vec{a} = (2, 1)$ and $\vec{b} = (-3, 4)$, then the projection vector of vector $\vec{a}$ on vector $\vec{b}$ is ()

- A. $(\frac{6}{5}, -\frac{8}{5})$ B. $(\frac{6}{25}, -\frac{8}{25})$ C. $(-\frac{6}{5}, \frac{8}{5})$ D. $(-\frac{6}{25}, \frac{8}{25})$

题目 6-4(Question 6-4)

已知 O 是坐标原点, $A(3,1)$, $B(-2,2)$, 则 $\overrightarrow{AB} + \overrightarrow{OA} = ()$

Given O is the origin, $A(3,1)$, $B(-2,2)$, then $\overrightarrow{AB} + \overrightarrow{OA} = ()$

A. $(-1,3)$ B. $(-2,2)$ C. $(-1,1)$ D. $(3,-1)$

题目 6-5(Question 6-5)

已知向量 $\vec{a} = (-1,1)$, $\vec{b} = (2,x)$, 若 $\vec{a} \parallel \vec{b}$, 则 $|\vec{a} - \vec{b}| = ()$

Given vectors $\vec{a} = (-1,1)$ and $\vec{b} = (2,x)$, if $\vec{a} \parallel \vec{b}$, then $|\vec{a} - \vec{b}| = ()$

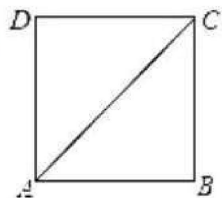
A. $3\sqrt{2}$ B. 3 C. $2\sqrt{2}$ D. 2

题型 7: 解决 $\overrightarrow{AB} \cdot \overrightarrow{AC}$ 的 3 种常用方法(Three Common Methods to Solve $\overrightarrow{AB} \cdot \overrightarrow{AC}$)

针对训练 7-1(Targeted Training 7-1)

如图正方形 $ABCD$ 的边长为 3, 求 $\overrightarrow{AB} \cdot \overrightarrow{AC} = \underline{\hspace{2cm}}$.

As shown in the figure, the side length of square $ABCD$ is 3, find $\overrightarrow{AB} \cdot \overrightarrow{AC} = \underline{\hspace{2cm}}$.



针对训练 7-2(Targeted Training 7-2)

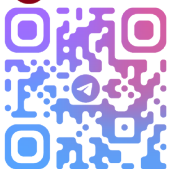
在 $\triangle ABC$ 中, M 是 BC 的中点, $AM = 3$, $BC = 10$, 则 $\overrightarrow{AB} \cdot \overrightarrow{AC} = \underline{\hspace{2cm}}$.

In $\triangle ABC$, M is the midpoint of BC , $AM = 3$, $BC = 10$, then $\overrightarrow{AB} \cdot \overrightarrow{AC} = \underline{\hspace{2cm}}$.

针对训练 7-3(Targeted Training 7-3)

在 $\triangle ABC$ 中, O 为中线 AM 上一个动点, 若 $AM = 2$, 则 $\overrightarrow{OA} \cdot (\overrightarrow{OB} + \overrightarrow{OC})$ 的最小值是 $\underline{\hspace{2cm}}$.

In $\triangle ABC$, O is a moving point on the median AM , if $AM = 2$, then the minimum value of $\overrightarrow{OA} \cdot (\overrightarrow{OB} + \overrightarrow{OC})$ is $\underline{\hspace{2cm}}$.



针对训练 7-4(Targeted Training 7-4)

已知 $\triangle ABC$ 是边长为 1 的等边三角形,点 D 、 E 分别是边 AB 、 BC 的中点,连接 DE 并延长到点 F ,使得 $DE = 2EF$,则 $\overrightarrow{AF} \cdot \overrightarrow{BC}$ 的值为 ()

Given $\triangle ABC$ is an equilateral triangle with side length 1, points D and E are the midpoints of sides AB and BC respectively. Connect DE and extend it to point F such that $DE = 2EF$, then the value of $\overrightarrow{AF} \cdot \overrightarrow{BC}$ is ()

A. $-\frac{5}{8}$ B. $\frac{1}{4}$ C. $\frac{1}{8}$ D. $\frac{11}{8}$

针对训练 7-5(Targeted Training 7-5)

正方形 $ABCD$ 的边长是 2, E 是 AB 的中点,则 $\overrightarrow{EC} \cdot \overrightarrow{ED} = ()$

The side length of square $ABCD$ is 2, E is the midpoint of AB , then $\overrightarrow{EC} \cdot \overrightarrow{ED} = ()$

A. $\sqrt{5}$ B. 3 C. $2\sqrt{5}$ D. 5

针对训练 7-6(Targeted Training 7-6)

在 $\triangle ABC$ 中, $\angle A = 90^\circ$, $AB = AC = 2$, 点 M 为边 AB 的中点,点 P 在边 BC 上,则 $\overrightarrow{MP} \cdot \overrightarrow{CP}$ 的最小值为_____.

In $\triangle ABC$, $\angle A = 90^\circ$, $AB = AC = 2$, M is the midpoint of side AB , P is on side BC , then the minimum value of $\overrightarrow{MP} \cdot \overrightarrow{CP}$ is _____.

针对训练 7-7(Targeted Training 7-7)

已知正方形 $ABCD$ 的边长为 2, 点 P 满足 $\overrightarrow{AP} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC})$, 则 $|\overrightarrow{PD}| =$ _____; $\overrightarrow{PB} \cdot \overrightarrow{PD} =$ _____.

Given the side length of square $ABCD$ is 2, point P satisfies $\overrightarrow{AP} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC})$, then $|\overrightarrow{PD}| =$ _____; $\overrightarrow{PB} \cdot \overrightarrow{PD} =$ _____.

针对训练 7-8(Targeted Training 7-8)

已知 $\overrightarrow{AB} = (2,3)$, $\overrightarrow{AC} = (3,t)$, $|\overrightarrow{BC}| = 1$,则 $\overrightarrow{AB} \cdot \overrightarrow{BC} = ()$

Given $\overrightarrow{AB} = (2,3)$, $\overrightarrow{AC} = (3,t)$, and $|\overrightarrow{BC}| = 1$, then $\overrightarrow{AB} \cdot \overrightarrow{BC} = ()$

A. -3 B. -2 C. 2 D. 3

针对训练 7-9(Targeted Training 7-9)

已知 $\triangle ABC$ 是边长为 2 的等边三角形, P 为平面 ABC 内一点,则 $\overrightarrow{PA} \cdot (\overrightarrow{PB} + \overrightarrow{PC})$ 的最小值是 $()$

Given $\triangle ABC$ is an equilateral triangle with side length 2, P is a point in the plane ABC , then the minimum value of $\overrightarrow{PA} \cdot (\overrightarrow{PB} + \overrightarrow{PC})$ is $()$

A. -2 B. $-\frac{3}{2}$ C. $-\frac{4}{3}$ D. -1

针对训练 7-10(Targeted Training 7-10)

线段 AB 的长度为 6, C 、 D 为其三等分点(C 靠近 A , D 靠近 B),若 P 为线段 AB 外一点,且满足 $\overrightarrow{PC} \cdot \overrightarrow{PD} = 0$,则 $\overrightarrow{PA} \cdot \overrightarrow{PB} = ()$

The length of segment AB is 6, C and D are its trisection points (C is close to A , D is close to B). If P is a point outside segment AB and satisfies $\overrightarrow{PC} \cdot \overrightarrow{PD} = 0$, then $\overrightarrow{PA} \cdot \overrightarrow{PB} = ()$

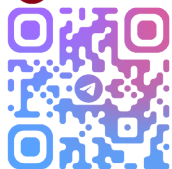
A. 36 B. -36 C. -8 D. 8

针对训练 7-11(Targeted Training 7-11)

已知长方形 $ABCD$ 的边 $AB = 4$, $AD = 2$, E 为 BC 的中点,则 $\overrightarrow{AE} \cdot \overrightarrow{BD} = ()$

Given the sides of rectangle $ABCD$ are $AB = 4$ and $AD = 2$, E is the midpoint of BC , then $\overrightarrow{AE} \cdot \overrightarrow{BD} = ()$

A. -14 B. 14 C. -18 D. 18



针对训练 7-12(Targeted Training 7-12)

已知 $\triangle ABC$ 的每条边长均为 2, D 、 E 分别是 BC 、 AC 的中点, 则 $\overrightarrow{DA} \cdot \overrightarrow{DE} = ()$

Given each side length of $\triangle ABC$ is 2, D and E are the midpoints of BC and AC respectively, then $\overrightarrow{DA} \cdot \overrightarrow{DE} = ()$

- A. $\frac{\sqrt{3}}{2}$ B. $\frac{3}{4}$ C. $\frac{3}{2}$ D. 3

针对训练 7-13(Targeted Training 7-13)

在 $\triangle ABC$ 中, $\cos A = \frac{1}{2}$, $AB = 2$, $AC = 3$, D 是线段 BC 靠近 C 的一个三等分点, 则 $\overrightarrow{AD} \cdot \overrightarrow{BD} = ()$

In $\triangle ABC$, $\cos A = \frac{1}{2}$, $AB = 2$, $AC = 3$, D is a trisection point of segment BC close to C , then $\overrightarrow{AD} \cdot \overrightarrow{BD} = ()$

- A. $\frac{22}{9}$ B. $-\frac{22}{9}$ C. $\frac{16}{9}$ D. $-\frac{8}{9}$

针对训练 7-14(Targeted Training 7-14)

三角形 ABC 中, D 是 BC 中点, $AB = 2$, $BC = 3$, $AC = 4$, 则 $\overrightarrow{AD} \cdot \overrightarrow{AB} = \underline{\hspace{2cm}}$.

In triangle ABC , D is the midpoint of BC , $AB = 2$, $BC = 3$, $AC = 4$, then $\overrightarrow{AD} \cdot \overrightarrow{AB} = \underline{\hspace{2cm}}$.

针对训练 7-15(Targeted Training 7-15)

已知 $\triangle ABC$ 中, $AB = AC = 5$, $BC = 6$, 点 D 为 AC 的中点, 点 M 为边 BC 上一动点, 则 $\overrightarrow{MD} \cdot \overrightarrow{MC}$ 的最小值为 $()$

Given $\triangle ABC$ with $AB = AC = 5$, $BC = 6$, D is the midpoint of AC , M is a moving point on side BC , then the minimum value of $\overrightarrow{MD} \cdot \overrightarrow{MC}$ is $()$

- A. 27 B. 0 C. $-\frac{7}{16}$ D. $-\frac{9}{16}$

针对训练 7-16(Targeted Training 7-16)

在正方形 $ABCD$ 中, $AB = 4$, P 、 Q 分别为 BC 、 CD 的中点, $BM = 3MA$, 则 $\overrightarrow{PQ} \cdot \overrightarrow{PM} =$ ()

In square $ABCD$ with $AB = 4$, P and Q are the midpoints of BC and CD respectively, $BM = 3MA$, then $\overrightarrow{PQ} \cdot \overrightarrow{PM} =$ ()

A. 2 B. 1 C. 10 D. 4

针对训练 7-17(Targeted Training 7-17)

设四边形 $ABCD$ 为平行四边形, $|\overrightarrow{AB}| = 6$, $|\overrightarrow{AD}| = 4$, 若点 M 、 N 满足 $\overrightarrow{BM} = 3\overrightarrow{MC}$, $\overrightarrow{DN} = 2\overrightarrow{NC}$, 则 $\overrightarrow{AM} \cdot \overrightarrow{NM} =$ ()

Let quadrilateral $ABCD$ be a parallelogram, $|\overrightarrow{AB}| = 6$, $|\overrightarrow{AD}| = 4$. If points M and N satisfy $\overrightarrow{BM} = 3\overrightarrow{MC}$ and $\overrightarrow{DN} = 2\overrightarrow{NC}$, then $\overrightarrow{AM} \cdot \overrightarrow{NM} =$ ()

A. 20 B. 15 C. 9 D. 6

针对训练 7-18(Targeted Training 7-18)

在 $\triangle ABC$ 中, M 是 BC 的中点, $AM = 1$, 点 P 在 AM 上且满足 $\overrightarrow{AP} = 2\overrightarrow{PM}$, 则 $\overrightarrow{PA} \cdot (\overrightarrow{PB} + \overrightarrow{PC}) =$ ()

In $\triangle ABC$, M is the midpoint of BC , $AM = 1$, P is on AM and satisfies $\overrightarrow{AP} = 2\overrightarrow{PM}$, then $\overrightarrow{PA} \cdot (\overrightarrow{PB} + \overrightarrow{PC}) =$ ()

A. $-\frac{4}{9}$ B. $-\frac{4}{3}$ C. $\frac{4}{3}$ D. $\frac{4}{9}$

题型 8: 三点共线模型专练 $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AC}$ (Special Training on Collinear Three-Point Model $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AC}$)

题目 8-1(Question 8-1)

在 $\triangle ABC$ 中, M 是 AC 边上一点, 且 $\overrightarrow{AM} = 2\overrightarrow{MC}$, 若 $\overrightarrow{BM} = x\overrightarrow{BA} + y\overrightarrow{BC}$, 则 y 的值为 ()



In $\triangle ABC$, M is a point on side AC with $\overrightarrow{AM} = 2\overrightarrow{MC}$. If $\overrightarrow{BM} = x\overrightarrow{BA} + y\overrightarrow{BC}$, then the value of y is ()

- A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C. $-\frac{2}{3}$ D. $\frac{2}{3}$

题目 8-2(Question 8-2)

在 $\triangle ABC$ 中, 点 M 、 N 满足 $\overrightarrow{AM} = 2\overrightarrow{MC}$, $\overrightarrow{BN} = \overrightarrow{NC}$, 若 $\overrightarrow{MN} = x\overrightarrow{AB} + y\overrightarrow{AC}$, 则 $x =$ _____, $y =$ _____。

In $\triangle ABC$, points M and N satisfy $\overrightarrow{AM} = 2\overrightarrow{MC}$ and $\overrightarrow{BN} = \overrightarrow{NC}$. If $\overrightarrow{MN} = x\overrightarrow{AB} + y\overrightarrow{AC}$, then $x =$ _____, $y =$ _____。

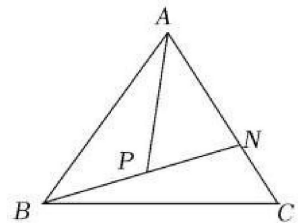
题目 8-3(Question 8-3)

如图, 在 $\triangle ABC$ 中, $\overrightarrow{AN} = 2\overrightarrow{NC}$, P 是 BN 上一点, 若 $\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$, 则实数 t 的值为 ()

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AN} = 2\overrightarrow{NC}$, P is a point on BN .

If $\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$, then the value of the real number t is ()

- A. $\frac{1}{6}$ B. $\frac{2}{3}$ C. $\frac{1}{2}$ D. $\frac{3}{4}$



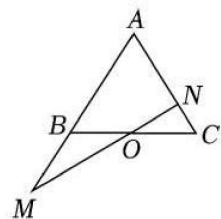
题目 8-4(Question 8-4)

如图所示, 在 $\triangle ABC$ 中, 点 O 是 BC 的中点, 过点 O 的直线分别交直线 AB 、 AC 于不同的两点 M 、 N , 若 $\overrightarrow{AB} = m\overrightarrow{AM}$, $\overrightarrow{AC} = n\overrightarrow{AN}$ ($m, n > 0$), 则 $m + n$ 的值为 ()

As shown in the figure, in $\triangle ABC$, point O is the midpoint of BC , and a straight line passing through O intersects straight lines AB and AC at

two distinct points M and N respectively. If $\overrightarrow{AB} = m\overrightarrow{AM}$ and $\overrightarrow{AC} = n\overrightarrow{AN}$ ($m, n > 0$), then the value of $m + n$ is ()

- A. 2 B. 3 C. $\frac{9}{2}$ D. 5

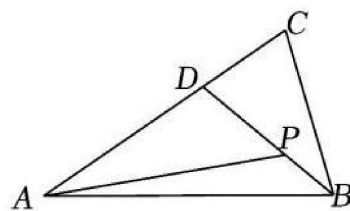


题目 8-5(Question 8-5)

如图,在 $\triangle ABC$ 中, $\overrightarrow{AD} = \frac{5}{8}\overrightarrow{AC}$, $\overrightarrow{BP} = \frac{2}{5}\overrightarrow{PD}$, 若 $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AC}$, 则 $\frac{\mu}{\lambda}$ 的值为 ()

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AD} = \frac{5}{8}\overrightarrow{AC}$ and $\overrightarrow{BP} = \frac{2}{5}\overrightarrow{PD}$. If $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AC}$, then the value of $\frac{\mu}{\lambda}$ is ()

- A. $\frac{2}{5}$ B. $\frac{5}{18}$ C. $\frac{5}{9}$ D. $\frac{18}{5}$

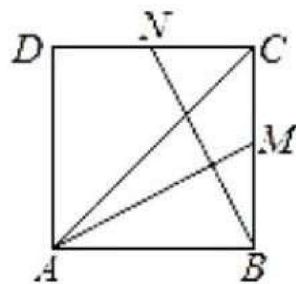


题目 8-6(Question 8-6)

如图,正方形 $ABCD$ 中, M 、 N 分别是 BC 、 CD 的中点,若 $\overrightarrow{AC} = \lambda\overrightarrow{AM} + \mu\overrightarrow{BN}$, 则 $\lambda + \mu =$ ()

As shown in the figure, in square $ABCD$, M and N are the midpoints of BC and CD respectively. If $\overrightarrow{AC} = \lambda\overrightarrow{AM} + \mu\overrightarrow{BN}$, then $\lambda + \mu =$ ()

- A. 2 B. $\frac{8}{3}$ C. $\frac{6}{5}$ D. $\frac{8}{5}$

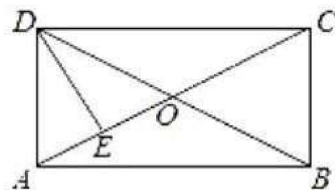


题目 8-7(Question 8-7)

如图所示,矩形 $ABCD$ 的对角线相交于点 O , E 为 AO 的中点,若 $\overrightarrow{DE} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$), 则 $\lambda \cdot \mu$ 等于 ()

As shown in the figure, the diagonals of rectangle $ABCD$ intersect at point O , and E is the midpoint of AO . If $\overrightarrow{DE} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$), then $\lambda \cdot \mu$ is equal to ()

- A. $-\frac{3}{16}$ B. $\frac{3}{16}$ C. $\frac{1}{2}$ D. $-\frac{1}{2}$

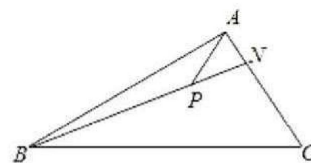




题目 8-8(Question 8-8)

如图,在 $\triangle ABC$ 中, $\overrightarrow{AN} = \frac{1}{3}\overrightarrow{NC}$, P 是 BN 上的一点,若 $\overrightarrow{AP} = m\overrightarrow{AB} + \frac{2}{9}\overrightarrow{AC}$,则实数 m 的值为 ()

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AN} = \frac{1}{3}\overrightarrow{NC}$, and P is a point on BN . If $\overrightarrow{AP} = m\overrightarrow{AB} + \frac{2}{9}\overrightarrow{AC}$, then the value of the real number m is



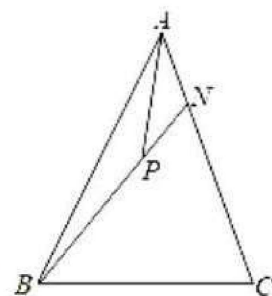
()

- A. $\frac{1}{9}$ B. $\frac{1}{3}$ C. 1 D. 3

题目 8-9(Question 8-9)

如图,在 $\triangle ABC$ 中, $\overrightarrow{AN} = \frac{2}{3}\overrightarrow{NC}$, P 是 BN 上一点,若 $\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$,则实数 t 的值为 ()

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AN} = \frac{2}{3}\overrightarrow{NC}$, and P is a point on BN . If $\overrightarrow{AP} = t\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}$, then the value of the real number t is ()



- A. $\frac{2}{3}$ B. $\frac{2}{5}$ C. $\frac{1}{6}$ D. $\frac{3}{4}$

题目 8-10(Question 8-10)

在矩形 $ABCD$ 中, $AB = 1$, $AD = \sqrt{3}$, P 为矩形内一点,且 $AP = \frac{\sqrt{3}}{2}$,若 $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$),则 $\lambda + \sqrt{3}\mu$ 的最大值为 ()

In rectangle $ABCD$, $AB = 1$, $AD = \sqrt{3}$, P is a point inside the rectangle, and $AP = \frac{\sqrt{3}}{2}$. If $\overrightarrow{AP} = \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ ($\lambda, \mu \in \mathbb{R}$), then the maximum value of $\lambda + \sqrt{3}\mu$ is ()

- A. $\frac{3}{2}$ B. $\frac{3+\sqrt{3}}{4}$ C. $\frac{\sqrt{6}}{2}$ D. $\frac{\sqrt{6}+3\sqrt{2}}{4}$

题目 8-11(Question 8-11)

已知点 P 在 $\triangle ABC$ 所在平面内,且 $\overrightarrow{PA} + 2\overrightarrow{PB} + 3\overrightarrow{PC} = \vec{0}$,设 $\overrightarrow{PD} = \overrightarrow{PA} + 2\overrightarrow{PB}$,则 $\frac{S_{\triangle PAB}}{S_{\triangle ACD}}$ 的值为 ()

It is known that point P is in the plane of $\triangle ABC$, and $\overrightarrow{PA} + 2\overrightarrow{PB} + 3\overrightarrow{PC} = \vec{0}$. Let $\overrightarrow{PD} = \overrightarrow{PA} +$

$\overrightarrow{PB}$, then the value of $\frac{S_{\triangle PAB}}{S_{\triangle ACD}}$ is ()

- A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{1}{5}$

题目 8-12(Question 8-12)

设 D 、 E 分别是 $\triangle ABC$ 的边 AB 、 BC 上的点, $AD = \frac{1}{2}AB$, $BE = \frac{2}{3}BC$, 若 $\overrightarrow{DE} = \lambda_1 \overrightarrow{AB} + \lambda_2 \overrightarrow{AC}$ (λ_1, λ_2 为实数), 则 $\lambda_1 + \lambda_2$ 的值为_____。

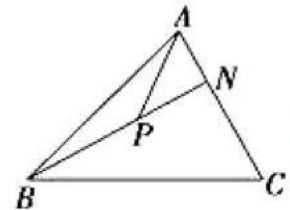
Let D and E be points on sides AB and BC of $\triangle ABC$ respectively, $AD = \frac{1}{2}AB$, $BE = \frac{2}{3}BC$. If $\overrightarrow{DE} = \lambda_1 \overrightarrow{AB} + \lambda_2 \overrightarrow{AC}$ (λ_1, λ_2 are real numbers), then the value of $\lambda_1 + \lambda_2$ is _____。

题目 8-13(Question 8-13)

如图所示, 在 $\triangle ABC$ 中, $\overrightarrow{AN} = \frac{1}{3}\overrightarrow{NC}$, P 是 BN 上的一点, 若

$\overrightarrow{AP} = m\overrightarrow{AB} + \frac{2}{11}\overrightarrow{AC}$, 则实数 $m =$ _____。

As shown in the figure, in $\triangle ABC$, $\overrightarrow{AN} = \frac{1}{3}\overrightarrow{NC}$, and P is a point on BN . If $\overrightarrow{AP} = m\overrightarrow{AB} + \frac{2}{11}\overrightarrow{AC}$, then the real number $m =$ _____。



题目 8-14(Question 8-14)

已知 $\vec{a}$, $\vec{b}$ 是不共线的向量, $\overrightarrow{OA} = \lambda\vec{a} + \mu\vec{b}$, $\overrightarrow{OB} = 3\vec{a} + 2\vec{b}$, $\overrightarrow{OC} = 2\vec{a} + 3\vec{b}$ 。若 A , B , C 三点共线, 则实数 λ , μ 满足 ()

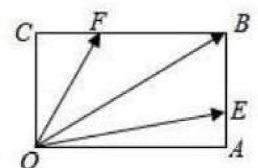
Given that $\vec{a}$ and $\vec{b}$ are non-collinear vectors, $\overrightarrow{OA} = \lambda\vec{a} + \mu\vec{b}$, $\overrightarrow{OB} = 3\vec{a} + 2\vec{b}$, $\overrightarrow{OC} = 2\vec{a} + 3\vec{b}$. If points A , B , C are collinear, then the real numbers λ and μ satisfy ()

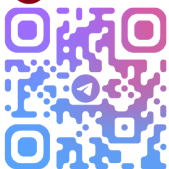
- A. $\lambda = \mu - 1$ B. $\lambda = \mu + 5$ C. $\lambda = 5 - \mu$ D. $\lambda = \mu + 1$

题目 8-15(Question 8-15)

如图, 在矩形 $OABC$ 中, 点 E , F 分别在 AB , BC 上, 且满足 $AB = 3AE$, $BC = 3CF$, 若 $\overrightarrow{OB} = \lambda\overrightarrow{OE} + \mu\overrightarrow{OF}$ ($\lambda, \mu \in \mathbb{R}$), 则 $\lambda + \mu =$ _____。

As shown in the figure, in rectangle $OABC$, points E and F are on AB





and BC respectively, satisfying $AB = 3AE$ and $BC = 3CF$. If $\overrightarrow{OB} = \lambda \overrightarrow{OE} + \mu \overrightarrow{OF}$ ($\lambda, \mu \in \mathbb{R}$),

then $\lambda + \mu =$ _____.